Random cross-validation understates cross-cohort calibration error in multi-cohort proteomics: an accession-aligned evaluation-protocol contrast

Zeyu Fu*,†,‡

†*State Key Laboratory of Trauma and Chemical Poisoning, Institute of Combined Injury, Chongqing Engineering Research Center for Nanomedicine, College of Preventive Medicine, Army Medical University, Chongqing, China*

‡*ORCID 0009-0001-8329-0108*

E-mail: fuzeyu99@126.com

Abstract

Cancer type and study are confounded in the six-cohort proteomics design examined here, so a protocol contrast cannot identify biological and acquisition effects separately. We audited tumor-versus-source-defined-normal classification in 1,191 CPTAC samples from 719 patients using 8,906 accession-aligned protein features. The shared feature intersection was defined using all six cohorts before evaluation, so LOGO is conditional on target-cohort feature availability rather than target-blind panel selection. Correcting cored-normal labels and patient identities, leave-one-cohort-out (LOGO) evaluation yielded equal-cohort macro AUROC 0.9343 and separately computed pooled AUROC 0.9555, with pooled ten-bin expected calibration error (ECE) 0.1100. Sample-stratified and true-patient-grouped five-fold cross-validation gave pooled AUROC 0.9942 and 0.9944, and ECE 0.0199 and 0.0135. Patient grouping did not erase the protocol gap. Missingness and three reference-normalized global summaries gave macro/pooled AUROC 0.5445/0.5116 and 0.8172/0.8492; differences from abundance are predictive increments, not biological signal fractions. Confidence-ranked abstention increased retained accuracy from 0.8497 to 119/119 at 10% coverage, with 107 retained samples from two easier cohorts. Model and recalibration sensitivities showed cohort-dependent effects despite

small pooled differences. The corrected evidence supports cohort-resolved evaluation of ranking, probability error and coverage in this design, not a deconfounded biological transfer estimate or clinical biomarker.

Keywords

proteomics; mass spectrometry; machine learning; cross-study validation; model calibration; data leakage; CPTAC; selective prediction

Introduction

This design cannot separate cancer-type biology from study-specific acquisition conditions. Each leave-one-cohort-out group is a distinct cancer/study cohort; the data do not establish that each cohort corresponds to one instrument, laboratory or acquisition run. Transfer discrimination and calibration therefore cannot be attributed to biology or technical effects alone[1]. The result is a protocol contrast under that confound, not a proteomic transfer law.

Quantitative mass-spectrometry proteomics now supports supervised classifiers for tumor status, subtype, and related endpoints, and the Clinical Proteomic Tumor Analysis Consortium (CPTAC) has made harmonized proteomic data for many cancer types openly available[2,3]. When such a classifier is reported, its performance is most often estimated by random k-fold cross-validation. In a multi-cohort proteomic dataset, that protocol places samples from the same study cohort on both sides of the train/test split. Such a split answers an interpolation question; presenting it as an estimate of transfer to an unseen study can give an optimistic impression of generalization[4]. The practically relevant question is rarely "how well does the model do on held-out samples from the studies it was trained on" but "how well, and how reliably, does it transfer to a new study" — which only a grouped, leave-one-study-out split estimates, and even then only as a protocol contrast when cancer type and study are locked together. This inflation of within-study, random-split performance relative to cross-study evaluation is not new to proteomics: cross-study validation frameworks for genomic and

transcriptomic prediction algorithms have previously shown that random cross-validation within a pooled multi-study dataset systematically overestimates performance relative to leave-one-study-out evaluation[5]. Our contribution is to demonstrate this protocol difference on real proteomics with a coupled discrimination-and-calibration analysis, not to report a new biological law.

Related work separates several causes of unreliable validation. The pipeML preprint recomputes global features within folds and supports cohort-aware validation, including leave-one-dataset-out analysis[6]. Our audit addresses the consequence of choosing a sample-level versus cohort-level evaluation unit after defining a fixed feature matrix; it does not introduce fold-aware preprocessing or grouped validation. Austin and colleagues describe a different failure: single-observation leave-one-out aggregation can underestimate rank performance because training-fold label prevalence is negatively associated with the held-out label, motivating rebalanced cross-validation[7]. That mechanism is distinct from the optimistic random-fold versus cohort-transfer contrast examined here; we do not infer that every grouped validation design is unbiased. In LC–MS data, BERNN couples representation learning and batch-effect removal to classification, and shows that stronger batch mixing need not preserve predictive information[8]. Our contribution is a coupled audit of discrimination, calibration, coverage and retained-cohort composition on a fixed proteomics matrix, not a new batch-correction method.

Discrimination (the area under the ROC curve, AUROC) is one axis of reliability; calibration — whether a reported probability matches the observed frequency — is the other, and modern flexible classifiers are frequently miscalibrated[9]. For a reliability audit, a well-ranked but overconfident model is dangerous precisely when it is wrong, so calibration must be measured on the same transfer split as discrimination. We therefore evaluate, on real CPTAC proteomics under leave-one-cohort-out, three coupled questions: (i) how high-dimensional abundance compares with missingness and global abundance summaries; (ii) how calibration varies across held-out cohorts and whether overconfidence tracks transfer difficulty; and (iii) how much a random-split protocol misstates both relative to the grouped split. We use tumor-versus-normal as a deliberately high-effect, source-defined

label so that the reliability phenomena — not the difficulty of the biological task — are what is under study.

The evaluation unit is a cohort, a patient, or an individual sample depending on the question, and those units are not interchangeable. A sample-level split estimates interpolation among the cohorts represented on both sides of the split; a leave-one-cohort-out split withholds an entire study cohort. Because the six cohorts pair cancer type with study, neither split identifies a biological transfer effect by itself. We therefore treat the contrast as an audit of evaluation design and calibration, keeping the feature matrix and label definition fixed while explicitly varying the split rule.

Three controls clarify what the high-dimensional matrix can and cannot establish. A missingness-only feature set asks whether detection patterns alone carry the label. A small global-abundance summary asks how well a low-dimensional comparator predicts the same labels. Patient-grouped splitting asks whether repeated patient samples explain the difference between sample-level and cohort-level evaluation. These controls are descriptive safeguards: they expose alternative explanations for apparent transfer, but they do not turn a confounded cohort design into a deconfounded biological experiment.

Calibration is a separate question from ranking. A classifier can order tumour and non-malignant samples well and still assign probabilities that are too extreme on the cohorts where that ranking is weakest. Dropping low-confidence calls can then raise retained accuracy mainly by discarding difficult cohorts. Discrimination, probability error, coverage and the identity of the retained samples therefore have to be read together, on the same split. None of those readouts uncouples cancer type from acquisition batch.

Materials and Methods

Data and measurement identity

We used the CPTAC `umich` proteomics tables distributed by the `cptac` Python package[10]. The six retained cohorts have at least 15 tumor and 15 source-defined nonmalignant samples each:

clear-cell renal-cell carcinoma (CCRCC), lung adenocarcinoma (LUAD), uterine corpus endometrial carcinoma (UCEC), head-and-neck squamous-cell carcinoma (HNSCC), lung squamous-cell carcinoma (LSCC), and pancreatic ductal adenocarcinoma (PDAC). We did not reprocess raw mass-spectrometry files. The 1,191 samples comprise 689 tumors, 496 adjacent-normal samples and six HNSCC cored-normal samples. The latter have the suffix `.C`; both these samples and the `.N` adjacent-normal samples receive label 0, and unsuffixed tumor samples receive label 1. Normal means the source's nonmalignant tissue category, not a healthy donor. Table 1 gives the composition of every cohort.

Patient identities are cohort-scoped sample identifiers after removal of the applicable `.N` or HNSCC `.C` suffix. Unknown suffixes, duplicate sample IDs, or conflicting identities cause the analysis to stop. There are 719 patients: 470 with both tissue classes, 219 with tumor only and 30 with normal only. A patient can contribute more than two samples; these counts are not a claim of exactly one matched pair per patient. Earlier versions stripped only `.N`, misclassified the six `.C` samples, and counted 723 patient keys. All three protocols here use the corrected identities and labels.

The primary matrix contains 8,906 stable ENSP accessions present in all six cohorts, with a consistent gene annotation. We remove accession version suffixes only after testing stable-accession uniqueness within each cohort. For the retained intersection, full accession versions also agree across cohorts. We do not collapse multiple proteins into a gene name. An earlier gene-indexed intersection retained the first protein per gene: 1,234 of its 9,675 gene names selected different stable accessions across cohorts. As a mapping sensitivity, we retain the 8,441 historical gene names whose first stable accession is consistent in every cohort. This sensitivity is not independent validation or proof that all isoform measurement ambiguity has been removed. The intersection of six cohorts was defined before the outer folds were evaluated. Thus the held-out cohort contributed its *feature-availability metadata* (which accessions were measurable), though not its labels, to the fixed feature matrix. This is a conditional comparison on a shared assay space, not a target-blind deployment test in which the feature panel is locked before an unknown cohort arrives.

Abundances are the UMICH reference-intensity-subtracted quantities returned by the package, not raw total protein signal. We applied no additional cohort-level normalization or batch correction to the primary matrix. Missing abundance entries can have multiple mass-spectrometric causes[11]; they are handled by the training-fold imputation below. The accession map and sample/fold manifest are supplied with the reproducibility package.

Task, features, and model

The task is tumor versus source-defined nonmalignant tissue. The primary feature set is the accession-aligned abundance vector. Two comparator sets use the same samples and accession matrix: a binary missingness mask, and three per-sample global summaries (mean and median observed abundance, and detection fraction). These are reference-normalized abundance summaries, not measurements of raw total intensity. Their performance differences are predictive comparisons and cannot partition biological signal into global and protein-specific fractions.

The primary classifier is XGBoost[12], with 150 trees, maximum depth 4, learning rate 0.1, subsample 0.8, column subsample 0.3, histogram tree method, four threads and log-loss evaluation. No class weighting, early stopping or nested hyperparameter tuning was used. The historical hyperparameters were retained; `random_state` was not explicitly set in these XGBoost fits. The corrected analyses used `cptac` 1.5.14, XGBoost 3.3.0, NumPy 2.5.0, pandas 3.0.3 and scikit-learn 1.9.0. Labels, feature identity, grouped folds and parts of the software environment differ from the historical analysis, so the revision is not a controlled one-factor ablation of its previous numbers.

Evaluation and sensitivity protocols

Leave-one-cohort-out (LOGO) trains on five cohorts and tests on the sixth, giving one out-of-cohort prediction per sample. The other protocols are sample-stratified five-fold cross-validation (`StratifiedKFold`, `shuffle=True`, `random_state=28`) and true-patient-grouped five-fold cross-validation (`GroupKFold`, without shuffling). The corrected patient-fold manifest replaces, rather than replays, the invalid historical partition. Joining by cohort and sample ID verifies the stored fold

assignments. Patient-grouped CV and LOGO have zero true-patient crossings; sample CV assigns samples from 381 patients to more than one evaluation fold. This is a patient count across the entire partition, not the mean fraction of pairs split in one train/test fold.

For every primary fit, per-feature medians are estimated on its training fold, with an entirely missing training feature filled by zero. The same medians are then applied to the held-out fold. The three global summaries are computed within each sample before this imputation. All model comparisons hold the corrected samples and outer LOGO folds fixed. A deliberately leaky sensitivity uses medians of the entire feature matrix, including held-out feature values but not held-out labels; it is a diagnostic, not a recommended protocol. The linear-model sensitivity uses training-fold median imputation and `StandardScaler`, followed by L2 logistic regression (`lbfgs`, $C = 1$, `max_iter=500`). All six linear fits converged. The iteration limit is 500, as in the executed code, not 1000 as stated by a superseded method-lock writer.

Metrics, aggregation and conditional sensitivity

We report within-cohort area under the receiver operating characteristic curve (AUROC), ten-bin expected calibration error (ECE), Brier score, log-loss and accuracy. Equal-cohort macro metrics are the arithmetic mean of the six within-cohort metrics, including for sample and patient CV. They are not the mean of the five CV-fold metrics. Pooled metrics are computed separately after concatenating all 1,191 predictions. Pooled AUROC includes positive–negative pairs from different cohorts whose scores came from distinct fitted models. It is therefore not the average of six cohort-transfer AUROCs. Within-cohort sample/patient CV metrics likewise combine that cohort's out-of-fold predictions.

For an evaluated set of n predictions, p_i is the predicted tumor probability and $y_i \in \{0, 1\}$ is the source-defined tissue label. Probability bins S_b have width 0.1 on $[0, 1]$, closed on the left and open on the right except for the last bin. ECE is

$$\mathrm{ECE} = \sum_{b=1}^{10} \frac{|S_b|}{n} \left| \overline{p}_{S_b} - \overline{y}_{S_b} \right|. \tag{1}$$

The bars are the arithmetic means within each nonempty bin. Equation 1 is evaluated separately for each cohort or concatenated prediction set; empty bins contribute zero. ECE depends on binning and sample composition; poorer discrimination does not mathematically force overconfidence or an increase in ECE. We use the Brier score and log-loss as complementary probability-error summaries (SI Section S2 and Tables S1–S6).

For a conditional sensitivity analysis, we resample true patients with replacement within each fixed source cohort, carrying every patient's stored predictions in all protocols together (2,000 draws; seed 20260922). We report the 2.5th–97.5th percentiles of paired protocol contrasts. These are fixed-prediction mixture-sensitivity ranges, not confidence intervals for a completely refitted learning algorithm or a future study. The six cancer/study cohorts are not assumed to be random exchangeable draws from a population of studies; resampling them alone would not establish a generic cross-study confidence interval.

Selective prediction ranks the pooled LOGO samples by $|p - 0.5|$, retaining the most confident fraction, with hard label $\mathbf{1}[p \geq 0.5]$. Ties retain source-row order, fixed by the sample manifest. For requested coverage c, the retained count is $\lfloor 1191c \rfloor$. We report both retained accuracy and the cohort composition of that same set. This is retrospective evaluation of stored predictions, not validation of an operational abstention threshold.

Post-hoc recalibration

For each outer held-out cohort, we divide the other five cohorts into the first two and the remaining three in the recorded cohort order. We train on each group and score the other, then concatenate those inner held-out predictions. Training-fold imputation is repeated inside each inner fit. No outer test label enters this procedure. Temperature scaling fits $T \in [0.05, 20]$ by negative-log-likelihood minimization, and Platt scaling fits logistic regression to the clipped probability logits (clip $[10^{-7}, 1 - 10^{-7}]$, $C = 10^6$, maximum 2000 iterations). These maps are applied to the already-stored outer LOGO probabilities.

The inner predictors train on two or three cohorts while the outer predictor trains on five. The analysis therefore evaluates this specific split-half recalibration procedure, not a universal mechanism of calibration failure under transfer. Each held-out cohort has a different map; even when each within-cohort map preserves ranking, pooled ranking can change when those different maps are applied across cohorts.

Independent side-study and historical dependencies

A separate ovarian (OV) side-study uses one cancer type, 12 observed TMT plexes, and the package's Tumor/Normal annotation: 103 samples (84 tumor, 19 normal) and 10,589 protein features, with QC and reference samples excluded. The abundance–annotation join covers all 103 samples. Its leave-one-TMT-plex-out protocol (plex-LOGO) uses train-fold medians and the same XGBoost settings except for explicitly setting `random_state=28`. We retain its recorded result as a separately scoped historical boundary observation, not as a rerun on the corrected six-cohort matrix. The source hashes and annotation totals were checked during integration; full prediction-level replay of OV remains separate.

Historical cohort-identity, permutation, location/scale, residual, injection, higher-moment and PC-removal diagnostics use the superseded gene-indexed matrix. Their numerical claims are not retained as evidence for the corrected primary matrix. The BCM/UMICH three-cohort dual-platform analysis has a distinct gene mapping and patient join and also requires its own identity audit before quantitative reuse. The Supporting Information records the disposition of each dependency, so these archived experiments are not silently promoted into the new result.

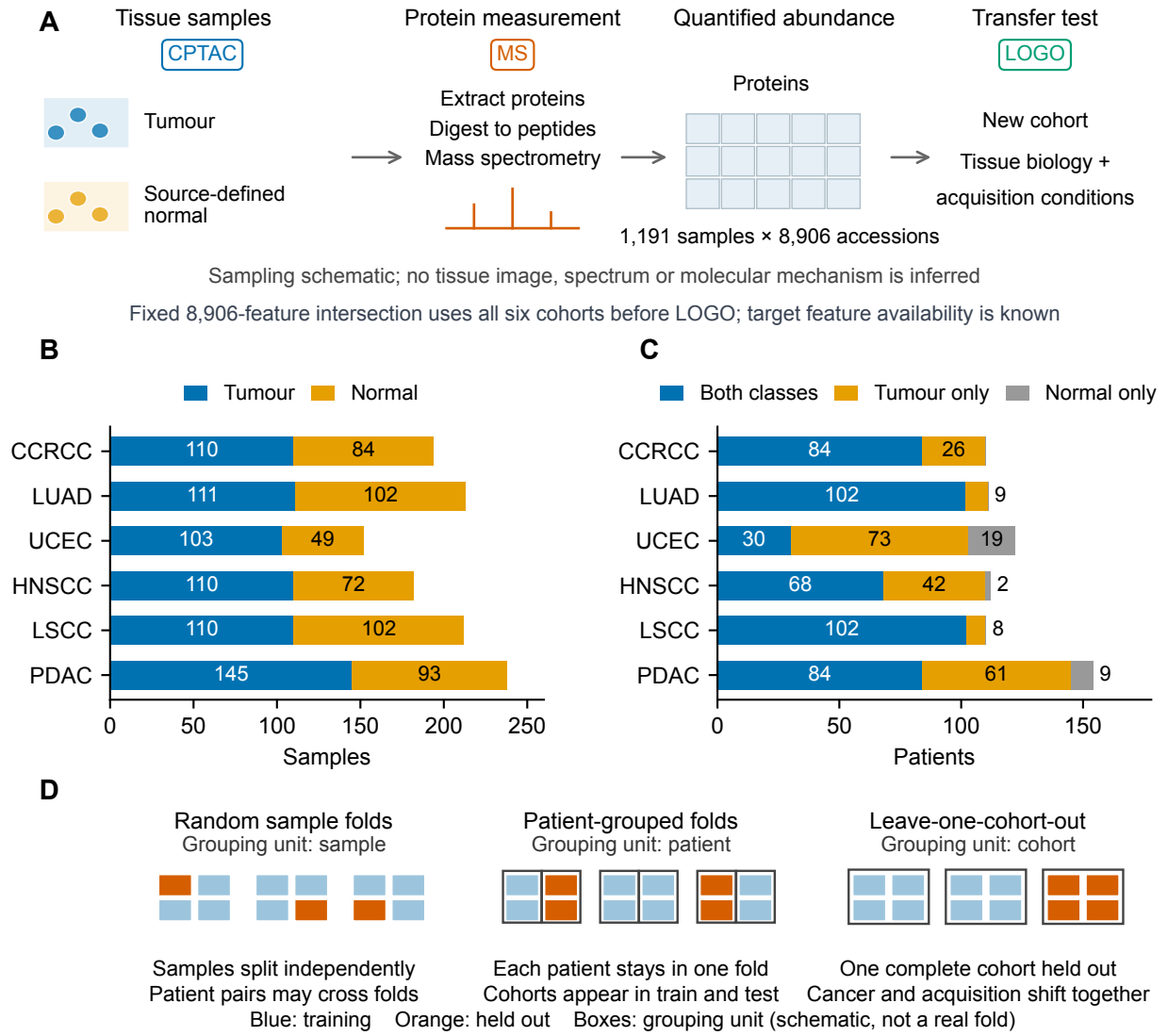


Figure 1. From tissue sampling to cohort transfer. (A) Conceptual path from tumor and source-defined nonmalignant tissue through mass spectrometry to the abundance matrix; tissue symbols and peaks are schematic, not observed histology or spectra. (B) Six cohorts contribute 1,191 samples: 689 tumors and 502 nonmalignant samples, including six HNSCC cored normals. (C) The 719 true patients include 470 with both tissue classes; some patients have more than two samples. (D) Sample folds may split a patient's samples, patient folds keep them together, and LOGO holds out an entire cancer/study cohort. The blocks illustrate grouping rules, not actual assignments. The 8,906-feature intersection uses all six cohorts before LOGO, so target feature availability is known. Cancer type and study are confounded; the data do not establish one instrument or one acquisition run per cohort.

Results

Accession-aligned abundance improves on descriptive comparators

The full-abundance LOGO classifier has equal-cohort macro AUROC 0.9343 and separately computed pooled AUROC 0.9555. These are different estimands: a cohort-average transfer result and a cross-model pooled ranking statistic. Macro and pooled ECE are 0.1193 and 0.1100, respectively. The within-cohort AUROCs range from 0.8485 (CCRCC) and 0.8610 (PDAC) through 0.9184 (UCEC) and 0.9809 (HNSCC) to 0.9974 (LSCC) and 0.9998 (LUAD) (Figure 2).

The missingness comparator has macro/pooled AUROC 0.5445/0.5116, while the three reference-normalized global summaries reach 0.8172/0.8492. Thus abundance performs better than either comparator at both levels of aggregation, but missingness is not exactly uninformative in every cohort. The high-dimensional model's increment over the three-summary model is 0.1172 macro or 0.1063 pooled (Figure 3). Neither increment is an identified contribution of protein-specific biology. AUROC differences do not decompose the sources of a signal, and these summaries are not raw signal-depth measurements. The first-accession-consistent-gene sensitivity has macro/pooled LOGO AUROC 0.9329/0.9557, close to the primary accession-aligned analysis (SI Table S4). This is a mapping robustness check on the same samples, not replication. The identity audit in SI Section S1 is part of the result boundary: the six HNSCC ".C" samples are source-defined cored normals, the corrected manifest contains 1,191 samples from 719 cohort-scoped patients, and the primary matrix contains 8,906 stable ENSP accessions. The retired 9,675-name first-protein mapping selected different stable accessions for 1,234 names; the 8,441-name analysis is therefore a consistent-accession sensitivity, not an independent cohort. Table 1 gives the corrected tissue composition, and Figure 1 links those samples to the three evaluation units. Neither source-defined normal tissue nor a schematic grouping rule is an independent clinical validation cohort.

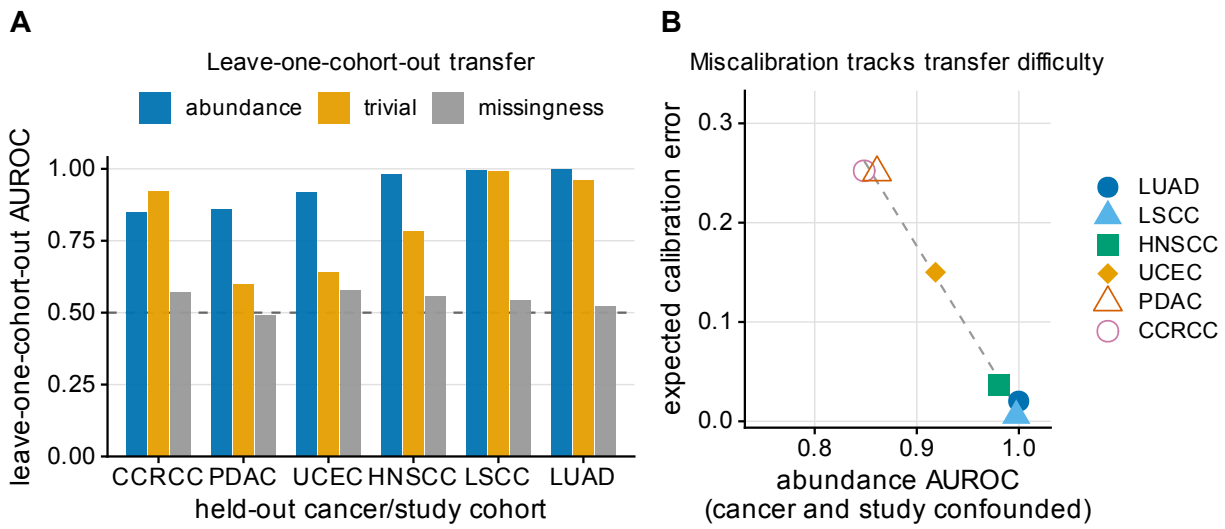


Figure 2. Heterogeneous transfer on the corrected six-cohort matrix. (A) Within-cohort LOGO AUROC for accession-aligned abundance (blue), three global summaries (orange) and missingness (grey). The dashed line marks chance ranking. (B) Each cohort's abundance AUROC and ten-bin ECE. Low-AUROC cohorts have larger calibration errors in this particular analysis, but no deterministic relation or causal mechanism is inferred. Each point is a cancer/study cohort; cancer type and study are confounded, and a cohort is not asserted to be a single acquisition run.

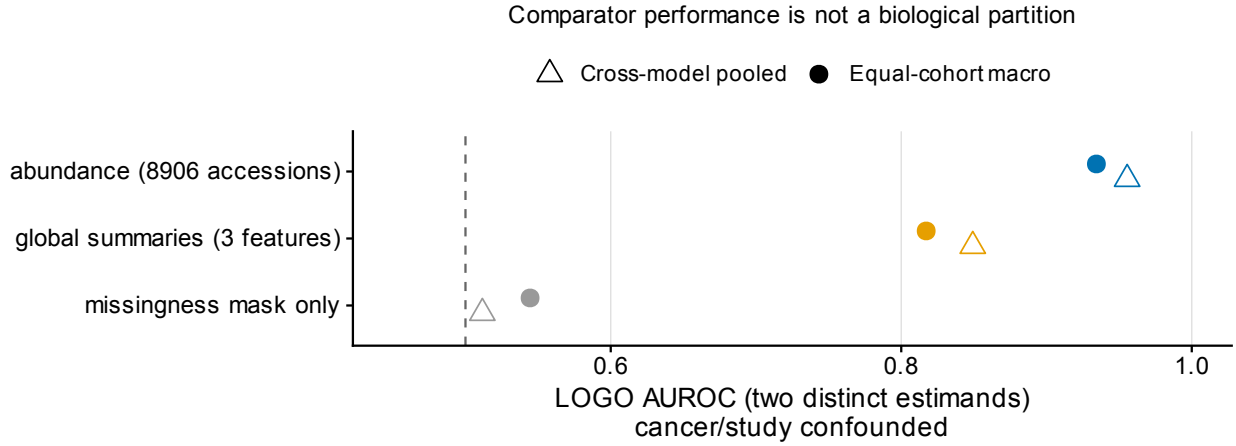


Figure 3. Comparator performance is not a biological signal partition. Filled circles show equal-cohort macro AUROC; open triangles show the separately computed pooled AUROC, which includes cross-cohort pairs of scores from different fitted models. Missingness, three global summaries and 8,906-accession abundance yield macro/pooled values 0.5445/0.5116, 0.8172/0.8492 and 0.9343/0.9555, respectively. No historical bootstrap whiskers or biological percentages are reused. Cancer type and study remain confounded.

Calibration error concentrates on harder held-out cohorts

The three highest-AUROC cohorts have ECE 0.0057 (LSCC), 0.0203 (LUAD) and 0.0368 (HNSCC). ECE is higher in UCEC (0.1501), PDAC (0.2506) and CCRCC (0.2522). The reliability curves and probability-error metrics indicate heterogeneous calibration, not uniform unreliability. An AUROC–ECE association in six selected cohorts does not show that poor discrimination forces confident errors. Nor does it distinguish tissue biology from acquisition differences. Brier scores and log-loss for the same prediction rows are reported in SI Tables S1–S3, with their definitions in SI Section S2. For abundance, CCRCC and PDAC have Brier scores 0.2245 and 0.2533 and log-loss 0.8162 and 0.7602, compared with 0.0064 and 0.0048, and 0.0253 and 0.0338, for LUAD and LSCC (SI Table S1). These complementary errors support the cohort-resolved description; their relation to ECE is not a mathematical necessity. All three tables use the same stored predictions, not additional prediction runs.

Random and patient-grouped CV differ from cohort transfer

Sample CV has macro/pooled AUROC 0.9909/0.9942, and true-patient-grouped CV has 0.9916/0.9944 (Table 2). Compared with LOGO, the sample-CV advantage is 0.0565 macro and 0.0386 pooled. Patient-CV advantages are 0.0572 macro and 0.0389 pooled. Sample minus patient CV is −0.0007 macro and −0.0003 pooled. Preventing patient crossings therefore does not erase the observed protocol gap in this matrix. This finding does not isolate a batch effect or establish that patient structure can never affect performance under another design.

Within fixed cohorts, paired resampling of patients' stored predictions gives a 0.0442–0.0703 percentile range for the macro sample-minus-LOGO contrast and 0.0305–0.0478 for its pooled counterpart. These conditional mixture-sensitivity ranges omit model-refit variability and future-study uncertainty. They cannot be substituted for a general cross-study CI. The paired patient draws and their shared protocol contrasts are documented in SI Table S7 and Section S5; the ranges are conditional on the corrected patient keys, not on the historical 723-key partition.

Table 1. Corrected patient and tissue composition. Nonmalignant samples include both adjacent-normal and HNSCC cored-normal samples. "Both" counts patients with at least one sample of each class, not one-to-one sample pairs. The six HNSCC cored samples are included in its 72 normals.

Cohort	Samples	Tumor/normal	Patients	Both
CCRCC	194	110/84	110	84
LUAD	213	111/102	111	102
UCEC	152	103/49	122	30
HNSCC	182	110/72	112	68
LSCC	212	110/102	110	102
PDAC	238	145/93	154	84
Total	1191	689/502	719	470

Table 2. Three protocols on the same corrected accession matrix. Macro is the equal-weight mean of six within-cohort metrics, not the mean of five CV-fold values. Pooled metrics concatenate all predictions. Sample CV splits 381 patients across folds; patient CV and LOGO split none. No protocol here identifies a causal acquisition-batch effect.

Protocol	Macro AUROC	Pooled AUROC	Macro ECE	Pooled ECE
Sample CV	0.9909	0.9942	0.0269	0.0199
Patient CV	0.9916	0.9944	0.0219	0.0135
LOGO	0.9343	0.9555	0.1193	0.1100

Pooled ECE is 0.0199 for sample CV and 0.0135 for patient CV, compared with 0.1100 for LOGO. The corresponding macro ECE values are 0.0269, 0.0219 and 0.1193. Figure 6 displays the reliability curves and pooled calibration errors. The fold scatter in Figure 4 shows that the difference also concerns heterogeneity: cohort holdout produces a wider spread of transfer performance. Those folds have overlapping training sets and are not independent statistical replicates.

Small pooled sensitivity differences can hide cohort changes

Full-dataset-median XGBoost has macro/pooled LOGO AUROC 0.9326/0.9543, compared with 0.9343/0.9555 for training-fold medians. Its pooled ECE is 0.1144, versus 0.1100. The pooled AUROC change of −0.00125 is small, but CCRCC changes by +0.00249 and UCEC by −0.00693. This diagnostic does not justify leaking held-out feature values or claim that imputation is immaterial in other datasets.

Training-fold-imputed and standardized L2 logistic regression has macro/pooled AUROC 0.9341/0.9556 and pooled ECE 0.0797. Its pooled AUROC is only 0.00010 higher than XGBoost,

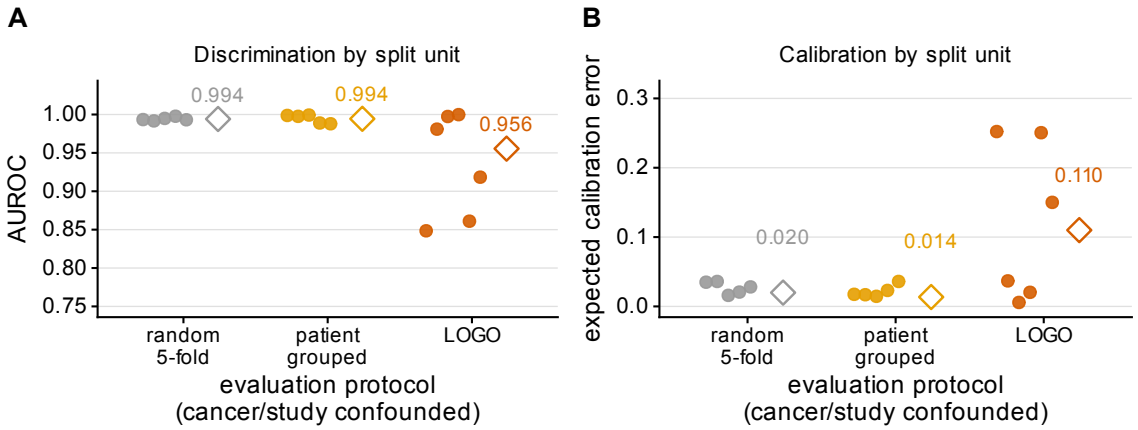


Figure 4. Performance depends on the held-out unit. Filled points show five sample-CV folds, five true-patient folds or six LOGO cohorts; open diamonds show separately computed pooled metrics. (A) AUROC. (B) Ten-bin ECE. The pooled values are not averages of the individual fold values. Fold scatter is descriptive, not a confidence interval; training sets overlap. Cancer type and study are confounded.

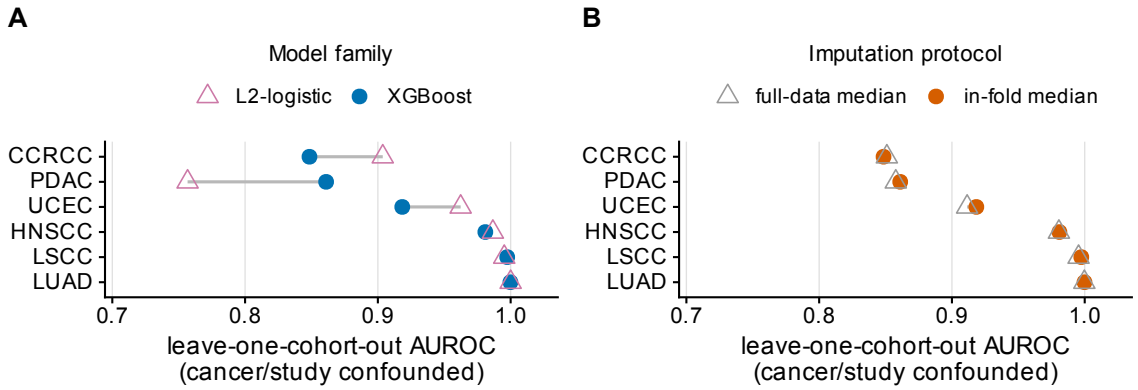


Figure 5. Cohort-resolved model and imputation sensitivities. (A) XGBoost versus L2 logistic regression, with training-fold imputation in both and training-fold standardization for L2. CCRCC AUROC changes from 0.8485 to 0.9037 and PDAC from 0.8610 to 0.7567. (B) Training-fold versus deliberately full-dataset medians for XGBoost, using identical outer folds. The same AUROC scale is used in both panels. No interval or significance claim is assigned to these single-fold comparisons. Cancer type and study are confounded.

yet CCRCC improves by 0.05520 and PDAC declines by 0.10434 (Figure 5). Pooled agreement therefore does not imply model-family invariance. These are matched-cohort sensitivity comparisons, without a claim of statistical significance for any single cohort's change. SI Tables S4–S6 retain the cohort-resolved mapping, L2 and deliberately leaky-median outcomes under the definitions in SI Section S4. All six L2 LOGO fits converged in 12–44 iterations under the executed 500-iteration limit (SI Table S8 and Section S6); convergence does not establish predictive superiority.

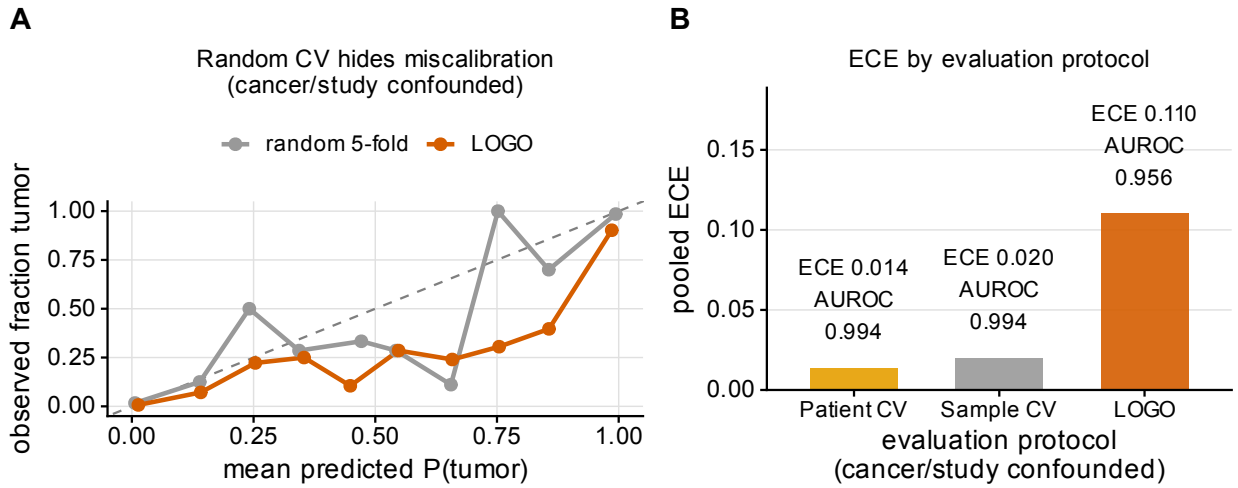


Figure 6. Cohort transfer has greater calibration error than cohort-mixing CV. (A) Mean probability versus observed tumor fraction in ten equal-width bins; the dashed line denotes perfect calibration. (B) Pooled ECE for sample CV (0.0199), patient CV (0.0135) and LOGO (0.1100), with pooled AUROC also shown. ECE uses each bin's sample mass, not equal bin weights. This is a protocol contrast among confounded cancer/study cohorts.

Confidence-ranked abstention changes cohort composition

Ranking pooled LOGO predictions by confidence raises retained accuracy from 0.8497 at full coverage to 119/119 at 10% requested coverage (Figure 7). The top-confidence set contains 52 LUAD, 55 LSCC, six CCRCC, four HNSCC, two PDAC and no UCEC samples. LUAD and LSCC together comprise 107/119 retained cases. The gain is therefore accompanied by strong selection of easier cohorts, but the composition alone does not identify how much improvement comes from between-cohort selection versus within-cohort case ranking. In particular, PDAC is not entirely absent from the retained set, contrary to the historical result. The complete composition is given in Table 3.

Selective classification[13] and calibration answer different questions. This retrospective curve does not repair probability calibration, validate a deployment threshold, or ensure reliability on a new PDAC- or UCEC-like population. Neither 119/119 correct retained predictions nor their pooled composition supplies a clinical performance guarantee.

Table 3. Cohort fractions in the confidence-ranked retained set. Rows use the same stored LOGO predictions as the accuracy curve. Requested coverage is rounded down to an integer retained count.

Coverage	n	CCRCC	LUAD	UCEC	HNSCC	LSCC	PDAC
0.1	119	0.050	0.437	0.000	0.034	0.462	0.017
0.2	238	0.050	0.378	0.008	0.059	0.466	0.038
0.3	357	0.048	0.322	0.031	0.109	0.431	0.059
0.4	476	0.061	0.305	0.057	0.126	0.382	0.069
0.5	595	0.082	0.277	0.079	0.148	0.331	0.082
0.6	714	0.090	0.258	0.104	0.155	0.284	0.109
0.7	833	0.104	0.229	0.124	0.168	0.250	0.125
0.8	952	0.114	0.211	0.131	0.167	0.222	0.154
0.9	1071	0.138	0.193	0.130	0.161	0.198	0.180
1.0	1191	0.163	0.179	0.128	0.153	0.178	0.200

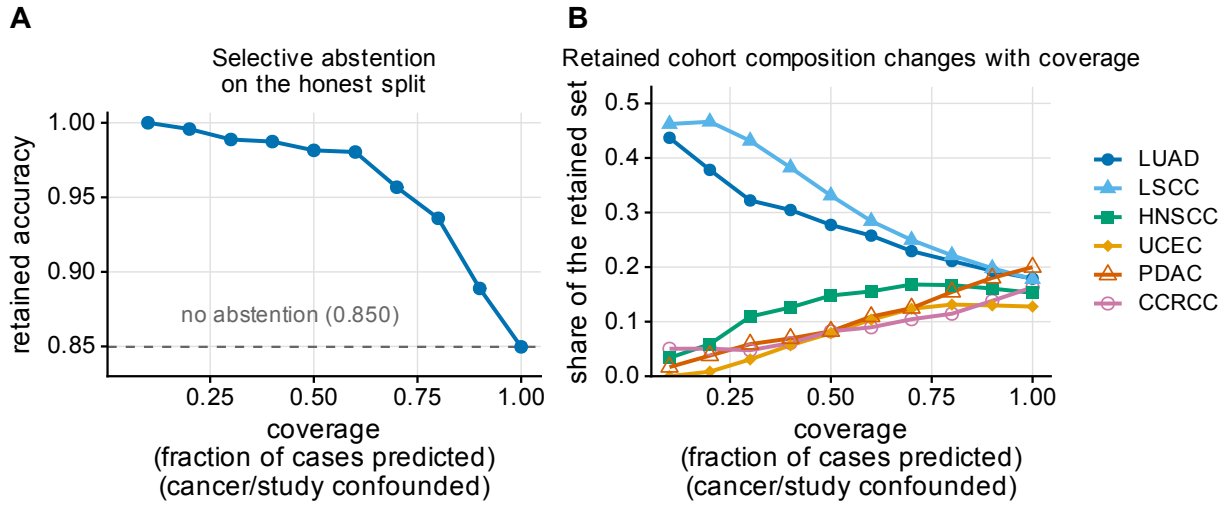


Figure 7. Retained accuracy and retained populations change together. (A) Accuracy among the highest-confidence LOGO predictions; the dashed line is full-coverage accuracy 0.8497. (B) Cohort fractions in the same sets. At 10% coverage, LUAD and LSCC account for 52/119 and 55/119 samples; PDAC contributes 2/119 and UCEC contributes none. The curves do not isolate within-cohort triage effects or validate an operational abstention rule. Cancer type and study are confounded.

A specific recalibration procedure has cohort-dependent effects

On the corrected outer LOGO predictions, split-half temperature scaling reduces pooled ECE from 0.1100 to 0.0845 and Platt scaling to 0.0388. Corresponding pooled Brier scores are 0.1135 before calibration, 0.0998 after temperature scaling and 0.0839 after Platt scaling. The cohort breakdown in Table 4 shows uneven effects: for example, Platt ECE falls from 0.2522 to 0.1303 on CCRCC but rises from 0.0203 to 0.0272 on LUAD.

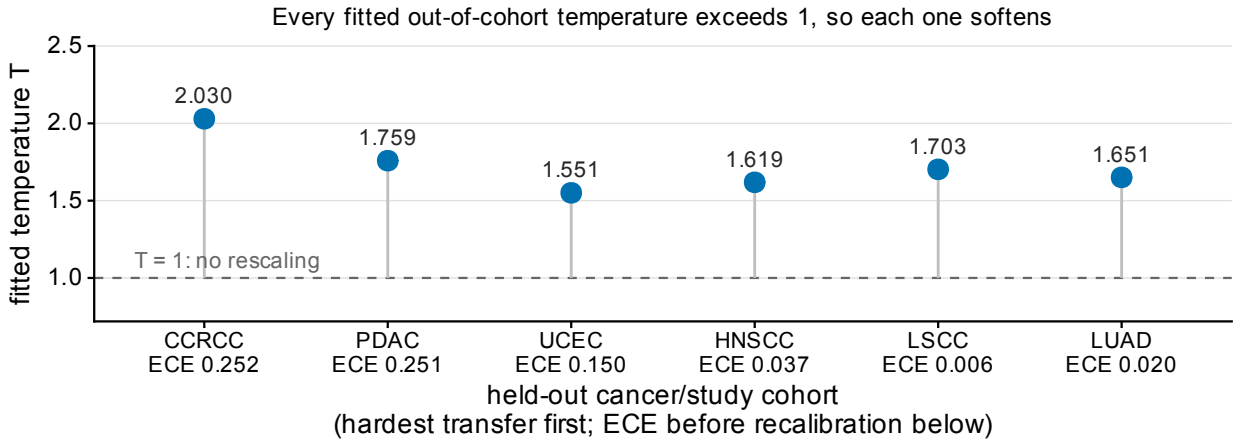


Figure 8. Temperatures fitted without outer held-out labels. All six estimated temperatures exceed one in this analysis; the dashed line denotes no rescaling. The printed ECE values are before recalibration. The calibration predictors trained on two or three cohorts, while the outer predictor trained on five. These maps therefore evaluate one specified inner/outer procedure. Per-cohort outcomes and the Platt alternative are in Table 4. Cancer type and study are confounded.

Table 4. Ten-bin ECE for the same outer LOGO predictions before and after the specified split-half recalibration procedure. Each cohort has its own map fitted using other cohorts only; pooled ECE is recomputed after concatenation, not averaged over these rows.

Cohort	Before	Temperature	Platt
CCRCC	0.2522	0.1844	0.1303
LUAD	0.0203	0.0463	0.0272
UCEC	0.1501	0.1401	0.0917
HNSCC	0.0368	0.0614	0.0370
LSCC	0.0057	0.0188	0.0063
PDAC	0.2506	0.1999	0.1408
POOLED	0.1100	0.0845	0.0388

The fitted temperatures range from 1.5506 to 2.0296 and all soften the probabilities (Figure 8). This describes these six fitted maps, not an intrinsic restriction of temperature scaling, which can also sharpen probabilities when $T < 1$. The calibration fit uses inner two-/three-cohort predictors rather than the outer five-cohort training regime. Consequently, heterogeneous benefit is a result for this procedure, not proof that a single recalibration method must fail under every transfer design. No held-out label was used to fit the maps. The inner split construction and bounded temperature/Platt fits are specified in SI Section S6, so this is a defined recalibration procedure rather than an unspecified post-hoc correction.

What the protocol contrast does not identify

The study does not identify a biological or technical fraction of the observed transfer performance. Each primary group corresponds to a distinct cancer/study cohort; the source metadata do not establish that each is a single acquisition batch. An observed sample/patient-versus-cohort contrast can motivate choosing a relevant evaluation unit without identifying why an individual cohort is hard.

The separately recorded OV side-study is retained as historical context: its archived summary reports pooled plex-LOGO abundance AUROC 0.985, close to random five-fold AUROC 0.987, with plex-LOGO ECE 0.046 and global-summary AUROC 0.951. It has only 103 samples, including 19 normals distributed as one to three per plex. It is one cancer type, a different feature table and a different holdout unit. The reported near-zero protocol contrast is not a prediction-replayed counterexample or a deconfounding test of the six-cohort result. The current fixed six-cohort comparison does not establish a universal gap; that restriction does not depend on treating the OV summary as verified new evidence. These are retained historical summary values with checked source provenance, not newly replayed OV predictions.

Numerical claims from historical six-cohort identification diagnostics, synthetic shifts, representation transforms and the separate dual-platform mapping are not used to strengthen the corrected primary conclusion. Their assumptions and current evidence status are itemized in SI Section S7, with the reproducibility boundary in Section S8. In particular, a label-dependent residual construction would not be a deployment evaluation, and a two-way variance calculation on a confounded design would not by itself identify biological and acquisition contributions. No pathway-enrichment analysis was performed.

Discussion

The central interpretation is about evaluation units. A random split can be internally consistent and still answer a less demanding question than transfer to a new cancer/study cohort. In this corrected

six-cohort matrix, both random and true-patient-grouped CV yield higher discrimination and lower calibration error than cohort holdout. The common accession panel was defined with knowledge of which features occur in the held-out cohort. This is not label leakage, but the contrast conditions on target-cohort feature availability; an unseen cohort with a prospectively locked panel remains unevaluated here. Patient grouping closes the identified patient-identity defect without erasing the contrast. It does not separate cancer-type biology from study-specific acquisition conditions.

The distinction between macro and pooled performance is consequential. Pooled AUROC combines scores from different fitted models and cross-cohort positive–negative pairs; the macro result averages within-cohort ranking. Neither should be silently substituted for the other. Fixed-prediction patient resampling gives a useful sensitivity to the sampled patients within these six studies, but not a complete uncertainty analysis of refitted learning procedures or future cohorts.

Global abundance summaries and missingness controls constrain interpretation of the high-dimensional model without converting AUROC increments into biological shares. Correct accession identity is necessary for this comparison; choosing the first protein associated with each gene does not guarantee a shared protein feature across cohorts. The source-normalized measurement scale also prevents the three-summary model from being described as a raw total-signal assay.

The imputation and model-family sensitivities show why similar pooled performance is insufficient evidence of invariance. L2 and XGBoost have almost the same pooled AUROC, but differ substantially on CCRCC and PDAC. Calibration and selective prediction likewise require cohort-resolved reporting: improved retained accuracy is accompanied by a changed cohort mixture, and the specified recalibration maps help some cohorts more than others. Neither supplies a deployment guarantee.

The correction and evaluation questions should remain separate. BERNN uses unlabeled validation and test features during representation warm-up[8]; its reported protocol is not an interchangeable numerical baseline for the transfer fits here. We have not benchmarked BERNN or pipeML[6] head-to-head and make no superiority claim. A future harmonization comparison should state whether unlabeled target features are available and report calibration as well as discrimination.

Our rejection curves are not conformal prediction sets and have no finite-sample coverage guarantee under cohort shift.

Tumor-versus-normal is a high-effect testbed, not a clinically actionable biomarker task. Six selected cohorts from one CPTAC processing source do not establish multi-platform generality. The separate OV summary is historical context without prediction-level replay in this revision; it is not independent evidence for or against the six-cohort contrast. Historical diagnostics on the superseded feature matrix remain archived but cannot substitute for new evidence after identities change. Independent measurements crossing cancer type with acquisition environment, broader model comparisons, and full refit-based uncertainty would be needed for stronger transfer claims.

The practical recommendation is to align the evaluation unit with the intended use, report within-cohort discrimination, calibration and coverage together, and distinguish macro from pooled summaries. The corrected data still support a protocol contrast in this design. They do not establish a general biological transfer law or identify a causal batch effect.

Supporting Information

The accompanying Supporting Information gives corrected sample and feature definitions, cohort-resolved AUROC and probability errors, conditional resampling ranges, L2 convergence checks, and an explicit disposition of historical side studies. Its tables and the main-text tables are generated from the same verified prediction records.

Data Availability Statement

All biological inputs are public CPTAC tables accessed through `cptac`[10]. The primary analysis uses the `umich` source for CCRCC, LUAD, UCEC, HNSCC, LSCC and PDAC. No new biological data were generated.

The current code and research artifacts are available at https://github.com/PeterPonyu/b28-proteomics-transfer, release tag `author-review-2026-10-02`, with the version-

specific archive https://doi.org/10.5281/zenodo.23101188. The package contains this Article and Supporting Information, their source and figure inputs, corrected sample/patient/fold and accession maps, stored predictions, analysis and verification programs, software environment records and a SHA-256 inventory. Dated candidate filenames retain their provenance; the current public manifest binds the included files to this revision. Default offline checks verify stored predictions and metrics and render the supplied PDFs without fitting models or downloading assay data.

A complete scientific rerun requires separately acquired, matching CPTAC/UMICH abundance and mapping files and the recorded fit environment. `DATA_SOURCES.md` records cohort/source/loader versions, filenames and input hashes; unrecorded PDC study accessions and per-cohort version labels are explicitly identified as unknown. Third-party raw proteomics are not redistributed. Author-owned code is MIT licensed; authored text, figures and derived result tables are CC BY 4.0 to the extent of author rights. Upstream conditions remain applicable as detailed in `THIRD_PARTY_NOTICES.md`. The earlier archive https://doi.org/10.5281/zenodo.21870024 is historical, not the current corrected release. Artifact publication does not constitute independent scientific or clinical validation.

Author Information

Corresponding Author. Zeyu Fu — State Key Laboratory of Trauma and Chemical Poisoning, Institute of Combined Injury, Chongqing Engineering Research Center for Nanomedicine, College of Preventive Medicine, Army Medical University, Chongqing, China; ORCID 0009-0001-8329-0108; email fuzeyu99@126.com.

Notes. The author declares no competing financial interest. Funding: N/A — no external funding to declare. Acknowledgments: N/A.

References

(1) Leek, J. T.; Scharpf, R. B.; Bravo, H. C.; Simcha, D.; Langmead, B.; Johnson, W. E.; Geman, D.; Baggerly, K.; Irizarry, R. A. Tackling the Widespread and Critical Impact of Batch Effects in High-Throughput Data. *Nat. Rev. Genet.* **2010**, *11*, 733–739.

(2) Edwards, N. J.; Oberti, M.; Thangudu, R. R.; Cai, S.; McGarvey, P. B.; Jacob, S.; Madhavan, S.; Ketchum, K. A. The CPTAC Data Portal: A Resource for Cancer Proteomics Research. *J. Proteome Res.* **2015**, *14*, 2707–2713.

(3) Li, Y.; others Proteogenomic Data and Resources for Pan-Cancer Analysis. *Cancer Cell* **2023**, *41*, 1397–1406.

(4) Kapoor, S.; Narayanan, A. Leakage and the Reproducibility Crisis in Machine-Learning-Based Science. *Patterns* **2023**, *4*, 100804.

(5) Bernau, C.; Riester, M.; Boulesteix, A.-L.; Parmigiani, G.; Huttenhower, C.; Waldron, L.; Trippa, L. Cross-study validation for the assessment of prediction algorithms. *Bioinformatics* **2014**, *30*, i105–i112.

(6) Hurtado, M.; Pancaldi, V. A New Pipeline for Cross-Validation Fold-Aware Machine Learning Prediction of Clinical Outcomes Addresses Hidden Data-Leakage in Omics Based Predictors. *bioRxiv* **2026**, 2026.03.12.711429, Preprint.

(7) Austin, G. I.; Pe'er, I.; Korem, T. Distributional Bias Compromises Leave-One-Out Cross-Validation. *Sci. Adv.* **2025**, *11*, eadx6976.

(8) Pelletier, S. J.; Leclercq, M.; Roux-Dalvai, F.; de Geus, M. B.; Leslie, S.; Wang, W.; Lam, T. T.; Nairn, A. C.; Arnold, S. E.; Carlyle, B. C.; Precioso, F.; Droit, A. BERNN: Enhancing Classification of Liquid Chromatography Mass Spectrometry Data with Batch Effect Removal Neural Networks. *Nat. Commun.* **2024**, *15*, 3777.

(9) Guo, C.; Pleiss, G.; Sun, Y.; Weinberger, K. Q. On Calibration of Modern Neural Networks. Proceedings of the 34th International Conference on Machine Learning (ICML), PMLR 70. 2017; pp 1321–1330, arXiv:1706.04599.

(10) Lindgren, C. M.; Adams, D. W.; Kimball, B.; Boekweg, H.; Tayler, S.; Pugh, S. L.; Payne, S. H. Simplified and Unified Access to Cancer Proteogenomic Data. *J. Proteome Res.* **2021**, *20*, 1902–1910.

(11) Lazar, C.; Gatto, L.; Ferro, M.; Bruley, C.; Burger, T. Accounting for the Multiple Natures of Missing Values in Label-Free Quantitative Proteomics Data Sets. *J. Proteome Res.* **2016**, *15*, 1116–1125.

(12) Chen, T.; Guestrin, C. XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. 2016; pp 785–794.

(13) Geifman, Y.; El-Yaniv, R. Selective Classification for Deep Neural Networks. Advances in Neural Information Processing Systems (NeurIPS). 2017; arXiv:1705.08500.

TOC Graphic

